IT2011 - Artificial Intelligence and Machine Learning

Department of Information Technology, Faculty of Computing

Year 2 semester 1 (2026)

Tutorial 08

Unsupervised Learning Workshop 2

Hierarchical Clustering

You are given the following distance matrix between five clusters:

	A	B	C	D	E
A	0				
B	0.25	0			
C	0.40	0.30	0		
D	0.50	0.45	0.20	0	
E	0.60	0.35	0.55	0.25	0

i. Using the agglomerative hierarchical clustering approach with single linkage, perform the clustering process step by step.
   a) Identify the closest pair of clusters at each step and perform the merge.
   b) After each merge, update the distance matrix using the single linkage criterion.
   c) Clearly present the sequence (order) of cluster merges, including the corresponding distances.

ii. Based on the merging sequence obtained in part (b), construct a dendrogram. Clearly indicate the heights (distances) at which clusters are merged.

iii. Using the dendrogram:

a) Determine the number of clusters formed when the dendrogram is cut at a distance of 0.3.

b) Clearly list the clusters obtained at this level.